

Using Reward Shaping to Train Cognitive-based Control Policies for Intelligent Tutoring Systems

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Abstract—Intelligent tutoring systems are used to train humans by personalizing education systems. For conventional learning contexts such as mathematics or computer programming, agents in intelligent tutoring systems have been designed to respond to humans based on cognitive feedback. However, the same cannot be said for psycho-motor learning contexts. In this paper, we design and validate several model-based cognitive control policies that determine when to provide learners with automation assistance in a psychomotor task. The shaping of rewards used to train these policies is motivated by the important role of learners’ self-confidence while learning. The trained policies are implemented empirically in a user study utilizing a quadrotor landing simulator, and learners’ performance and self-confidence during the task are compared. The results show that the cognitive control objective of calibrating learners’ self-confidence to their performance leads to significantly better task performance than is achieved when the decision to provide automation assistance is driven solely by users’ performance. This motivates the need for continuing research in modeling of human cognitive factors and the design of appropriate control objectives grounded in literature on human cognition.

I. INTRODUCTION

Motivation and Problem Definition. Advances in technology and autonomous systems have decreased the required level of human expertise initially needed to operate these systems [1]. Despite this, there remains a need for human training when interacting with a variety of automated or autonomous technologies, such as personnel training or decision support tools in military applications [2], and AI designed for surgery or diagnosis assistance in the medical field [3], [4]. Often, conventional training may be defined in terms of a minimum number of hours of human experience with a system or skill, but intelligent tutoring or training systems (ITSs) can be designed to train humans more efficiently. ITSs are used to personalize education systems by closing the loop between the human model and the tutoring agent [5], [6], [7], [8], [9]. Many ITSs designed for conventional domains such as mathematics, language learning, or computer programming accomplish this by incorporating cognitive feedback into ITS design

[6]. In particular, self-confidence has been identified as a cognitive factor that motivates human behavior during learning as it plays an integral role in human understanding of achievements [10], [11], [12]. According to [13], self-confidence can be described as *one’s judgment about their own capability to accomplish a task or goal*. Furthermore, self-confidence is one of the important cognitive factors in designing effective human-automation interaction (HAI) through the modeling of human cognitive behavior [14], [15], [16]. However, even though self-confidence is recognized to be integral to learning, cognitive feedback has not been as widely incorporated into the design of ITSs for psychomotor tasks. Therefore, this paper focuses on developing a cognitive control strategy based on an empirically-driven model of human self-confidence to accelerate human learning in a simulated quadrotor landing task.

Gaps in Literature. There is growing interest in developing ITSs for psychomotor tasks, any motor action that is based on conscious, mental activity (e.g. driving a car) [17], [18], [19], [20]. However, unlike learning contexts where knowledge can be inferred from how ‘correct’ an answer is [21], assessing the ‘correctness’ of performance in a psychomotor task is more difficult. An alternative approach is to extrapolate the ‘correctness’ of an answer to the ‘correctness’ of the human’s self-assessment of their performance. For example, in prior work [22], the authors show that an automation assistance intervention based on the calibration of the human’s self-confidence to their performance leads to accelerated learning. While effective, this heuristic intervention policy could be improved by replacing it with an optimal or near-optimal ITS agent that chooses when to assist the human according to self-confidence state predictions. Bayesian modeling techniques have previously been used to model student learning behavior in in-classroom contexts [23], [21]. However, with respect to control policy design and synthesis, the question of how to design a reward that quantitatively represents multiple learning objectives from a tutor’s perspective, such as maximizing performance while maximizing the student’s self-efficacy to maintain motivation for learning, remains unanswered.

Contribution. In this paper, we present a model-based cognitive control strategy for improving learning outcomes in a psychomotor task. The novelty of our approach is in 1) the use of a Markov Decision Process (MDP) model to capture the state evolution of the human’s self-confidence and 2) reward shaping to design, synthesize, and validate different control policies that quantitatively capture well-established learning objectives from the education literature. We train

This material is based upon work supported by the National Science Foundation under Awards No. 1836952, 1836900, and 1652113. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

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a self-confidence MDP model using 99 sets of participant data and test four different optimal control policies each with 20 sets of participant data. Our results support the need for cognitive-state feedback to enhance human learning.

Outline. The paper is organized as follows. We describe the experiment design in Section II. Next, training and validation of the self-confidence model is discussed in Section III. In Section IV, four reward functions are used to train optimal policies using the model described in Section III. These policies are implemented experimentally and the results are presented and discussed in Sections V and VI, respectively. Concluding remarks and future work are described in Section VII.

II. EXPERIMENT DESIGN

We conduct a between-subjects user study in which participants practice landing a quadrotor in a desktop training simulator module, as shown in Figure 1. In the study, each participant’s goal is to learn how to successfully land a quadrotor manually within 20 trials using a keyboard.

A. Experiment and Controller Design

The experiment consists of 20 trials, in which the participant must land the quadrotor on a landing pad with appropriate bounds on the quadrotor speed ($|v| \leq 5 \frac{m}{s}$) and roll ($|\phi| \leq 10^\circ$) to ensure a safe landing. An unsafe landing occurs when the quadrotor lands on the landing pad but outside the safe landing conditions. An unsuccessful landing occurs when the quadrotor crashes outside the landing pad. After each trial, a mode selection policy determines if the next trial mode will be conducted in *manual* control mode (θ_{off}) or *shared* control mode (θ_{on}). In manual control mode, the user fully controls the quadrotor position and velocity. In shared control mode, the user’s input is augmented by a linear quadratic regulator that tracks a reference trajectory based on expert demonstrations using a linear time-invariant model of the quadrotor dynamics as well as a quadratic cost function that minimizes tracking error [24].

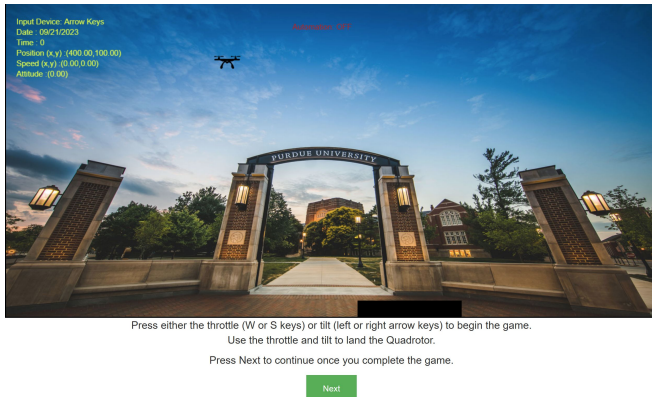


Fig. 1: Quadrotor experiment platform

B. Participants

In total, 180 participants completed the user study on Amazon Mechanical Turk [25]. Of these participants, 1 was removed due to restarting the study and completing more than 20 trials. Therefore, our data-set includes 179 participants (77 male and 102 female). Two implementations of the experiment were conducted. In the first, data from 99 participants was used for model training. In the second implementation, 20 participants were collected for each of the four different mode selection policies (described in more detail in Section IV). Participant ages range from 19–72 years ($\mu = 35.5$ years, $\sigma = 10.3$). All participants were compensated at a rate of \$8/hr. The Institutional Review Board at Purdue University approved the study.

C. Apparatus

The quadrotor landing training simulator was developed in Javascript and HTML. The simulator was originally developed for [24] and adapted for the current work. A keyboard is used to control the quadrotor.

D. Procedure.

Participants completed 20 trials of the quadrotor game. After every trial, participants are provided with their numerical score, the amount of time they expend in landing the quadrotor, and whether they *unsuccessfully*, *unsafely*, or *safely* land the quadrotor for all previous trials. Each participant is asked to rate their self-confidence in their ability to land the quadrotor on a numerical scale of 0-100. The following definition for self-confidence is provided to each participant: “*The confidence in oneself and one’s powers and abilities.*” In the first launch of the experiment, the control mode for each trial is randomized between manual and shared mode in order to achieve a rich data-set for training a self-confidence model. In the second launch of the game, the first two and last five trials are completed in manual mode so that each participant’s change in manual performance can be quantified. In trials 3–15, the participant’s control mode is determined by one of the four mode selection policies depending on the treatment to which they are assigned. Finally, after every trial k , the performance score S_k , time to land per trial, root mean square error between the participant’s trajectory and the expert reference trajectory, landing position coordinates, landing velocity, landing roll angle, control mode, and landing type (unsuccessful, unsafe, and safe) are recorded.

III. MDP MODELING OF SELF-CONFIDENCE

To construct a cognitive control strategy, we first train a data-driven model of human cognitive states. Similar to the author’s previous work on human trust and self-confidence modeling in [26] as well as work by [27] on modeling trust and workload, a probabilistic model of self-confidence is developed using an MDP. An MDP framework is chosen not only because it can capture the stochastic nature of human behavior, but it also allows for model-based reinforcement learning methods to be used for synthesizing optimal control

policies. An MDP is defined as a 5-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$ where $\mathcal{S}$ is a finite set of states and $\mathcal{A}$ is a finite set of actions. The transition probability function $\mathcal{T}$ is the probability of transitioning to the next state s' given the current state s and action a . The reward function $\mathcal{R}(s, a)$ represents the direct rewards associated with the agent's decisions depending on the state and actions. Finally, the discount factor γ is a value between 0 and 1 and represents how important future rewards are at the current state. The reward function and discount factor are used to train an optimal action (control) policy given the state dynamics. Here, an MDP is used as a gray-box modeling framework for the self-confidence model. This enables predictions of self-confidence to be parameterized by participant study data. In the self-confidence MDP, which we will call the SC MDP for brevity, the state is defined as the participant's *self-confidence* s_{SC} . The self-confidence state is discretized into low self-confidence SC_L , medium self-confidence SC_M , and high self-confidence SC_H . The 33% and 66% quantile of self-reported self-confidence values in the training dataset were used as thresholds to sort each trial into the discrete levels. The states are influenced by actions a in the set of actions $\mathcal{A}$. In this case, the actions are measures that influence self-confidence, such as the *performance* action a_P as well as the *automation* action a_θ which is an input to the task environment. The performance action a_P can take on levels of low P_L , medium P_M , and high P_H . Much like s_{SC} , the 33% and 66% quantile of all scores in the training dataset were used as thresholds to sort each trial into discrete levels of performance depending on the score each participant obtained in each trial. While performance could have been defined as a state of the model rather than an action, doing so would substantially increase the number of parameters in the SC MDP model, in turn making training more difficult. Given that the purpose of this model is for prediction of users' self-confidence, it was logical to define the states as such. The automation action a_θ represents if the automation assistance is off θ_{off} or on θ_{on} . The SC MDP model definition is summarized in Table I. It follows that the transition probabilities are given by $\mathcal{T}(s'|s, a) = \mathcal{T}(s_{SC}'|s_{SC}, a_P, a_\theta)$.

TABLE I: Definition of the human self-confidence (SC) MDP model.

States $s \in \mathcal{S}$ = [Self-Confidence s_{SC}]	$s_{SC} \in SC$:= $\left\{ \begin{array}{l} \text{Low Self-Confidence } SC_L \\ \text{Medium Self-Confidence } SC_M \\ \text{High Self-Confidence } SC_H \end{array} \right\}$
Actions $a \in \mathcal{A}$ = $\left[\begin{array}{l} \text{Performance } a_P \\ \text{Automation } a_\theta \end{array} \right]$	$a_P \in P$:= $\left\{ \begin{array}{l} \text{Low Performance } P_L \\ \text{Medium Performance } P_M \\ \text{High Performance } P_H \end{array} \right\}$
	$a_\theta \in \theta$:= $\left\{ \begin{array}{l} \text{Assistance off } \theta_{off} \\ \text{Assistance on } \theta_{on} \end{array} \right\}$

A. Training and Validation on Human Data

The genetic algorithm from MATLAB's Optimization Toolbox [28] is used to estimate the SC MDP model parameters by maximizing the likelihood of state sequences given the parameters. In prior work by the authors, the genetic algorithm was less sensitive to parameter initializing as well as achieved better log-likelihood values in comparison to other algorithms when training a coupled trust and self-confidence partially observable MDP [26]. The states and actions of all trials k are known, so the likelihood of state and action sequences is calculated using the joint probability of the series of states and actions up to trial k , i.e. $P(s_{0:k}, a_{1:k})$. The probability of each sequence can be calculated using the chain rule of probability and Markov property [29], [30], as in

$$P(s_{0:k}, a_{1:k}) = P(s_0) \times \prod_{i=1}^k P(s_i | a_i, s_{i-1}) \quad (1)$$

The likelihood of the state and action sequences until the end of the sequence at trial N is calculated by using transition probability matrices $\mathcal{T}(s'|s, a)$ in (1). The resulting transition probabilities are depicted in Table II.

The final model was validated to ensure that it was appropriate for optimal policy training. The resulting average negative log-likelihood of ten iterations of model training is $-1502.4 \pm 7.5790 \times 10^{-14}$. The standard error is $5.0446 \times 10^{-15}\%$ of the mean value, suggesting that the solution has indeed converged. To test for overfitting, a 5×2 -fold cross-validation is applied to the data in which the data is divided randomly into two equal sets or folds [31]. The average log-likelihood of the train and test sets from 5×2 -fold cross-validation is -745.91 ± 6.0094 and -1595.8 ± 12.399 , respectively.

To test the predictive capability of the model, multi-class Receiver operating characteristic (ROC) curves are used to illustrate the performance of the model in predicting the self-confidence cognitive state of each participant. The model is tested using data not used for model training. The ROC curves are generated by comparing the self-reported self-confidence to the predicted self-confidence. There are three discrete levels of self-confidence which serve as the different classes for the ROC curves. Three ROC curves are generated using a one-vs-rest scheme (Fig. 2b) and six ROC curves are generated using a one-vs-one scheme (Fig. 2a) [32]. In the one-vs-rest ROC, one class is treated as the positive class, and the rest are treated as negative classes; in the one-vs-one scheme, in every combination of two classes, data that is in any other class is removed, and an ROC curve is generated for binary classification using the remaining data. For each of these ROC curves, the true positive rate (TPR) and false positive rate (FPR) are plotted for threshold probabilities of 0-100% used to classify predicted self-confidence. The areas under the curve (AUC) can be used to infer model predictive capability. A classifier based on random guess has an AUC of 0.5. A perfect classifier will have an AUC of 1. Typically, an AUC between 0.7 and 0.8 and between

TABLE II: The self-confidence MDP model transition probabilities of transitioning to the next state $s' \in SC$ given the current state $s \in SC$, performance action $a_p \in P$ and automation assistance action $a_\theta \in \theta$.

	SC'_L	SC'_M	SC'_H
SC_L	0.63	0.34	0.032
SC_M	0.19	0.64	0.17
SC_H	0.10	0.30	0.60

(a) Low performance P_L and automation off θ_{off} .

	SC'_L	SC'_M	SC'_H
SC_L	0.71	0.23	0.059
SC_M	0.21	0.61	0.18
SC_H	0.14	0.26	0.60

(b) Medium performance P_M and automation off θ_{off} .

	SC'_L	SC'_M	SC'_H
SC_L	0.72	0.27	0.011
SC_M	0.16	0.60	0.24
SC_H	0.069	0.18	0.75

(c) High performance P_H and automation on θ_{off} .

	SC'_L	SC'_M	SC'_H
SC_L	0.55	0.40	0.050
SC_M	0.20	0.60	0.20
SC_H	0.064	0.31	0.63

(d) Low performance P_L and automation off θ_{on} .

	SC'_L	SC'_M	SC'_H
SC_L	0.69	0.245	0.062
SC_M	0.25	0.62	0.13
SC_H	0.089	0.26	0.65

(e) Medium performance P_M and automation off θ_{on} .

	SC'_L	SC'_M	SC'_H
SC_L	0.73	0.23	0.048
SC_M	0.17	0.63	0.21
SC_H	0.062	0.21	0.73

(f) High performance P_H and automation on θ_{on} .

0.8 and 0.9 is considered fair and good, respectively [33]. The ROC curves are generated using 80 sets of participant test data that was collected to test trained optimal policies later described in Section IV. Almost all the AUCs for each generated ROC curve are within the fair and good criteria. Therefore, the model is able to perform well when classifying self-confidence states, especially considering that the model is based on human cognitive state data.

IV. DESIGNING POLICY REWARDS FOR IMPROVED LEARNING

Next we design and synthesize four different optimal mode selection control policies using the trained SC MDP model, each based on a different reward function $R(s, a)$. We consider four different control objectives for the user: maximize performance, maximize self-confidence, calibrate self-confidence to performance, and maximize both performance and self-confidence. To facilitate comparison among the policies, the reward functions used to train the optimal policies are all designed such that all rewards range between 0 and 1. The optimal policies are trained from the reward functions using the Bellman optimality equations [29], [30]. Recall that the SC MDP model consists of two actions—the user’s performance and the automation mode—but only the automation mode a_θ is controllable. Therefore, when training the policies, the action-value function $Q(s, a)$ is modified such that the agent can only choose the optimal action pertaining to automation assistance a_θ . As a result, the new action-value function is calculated using

$$Q_\pi(s, a_p, a_\theta) = \mathcal{R}(s, a_p, a_\theta) + \gamma \sum_{\substack{s' \in \mathcal{S} \\ a_p \in P}} \mathcal{T}(s'|s, a_p, a_\theta) \sum_{a'_\theta \in \theta} Q(s', a'_p, a'_\theta). \quad (2)$$

When solving for the action-value function in matrix form, the set of transitions probabilities associated with each *controllable* automation action is represented by a $1 \times |\mathcal{S}||a_P|$ vector normalized to values of 0–1. A discount factor γ of

0.9 is used when training the policies. The reward functions $R_i(s, a)$ and resulting action-value functions $Q_i(s, a)$ for $i = 1, 2, 3, 4$ are shown in Tables III IV, V and VI, respectively. In each table, the columns represent the combinations of uncontrollable performance actions a_p and states, and the rows represent the controllable automation assistance actions a_θ .

A. Objective 1: Maximize Performance

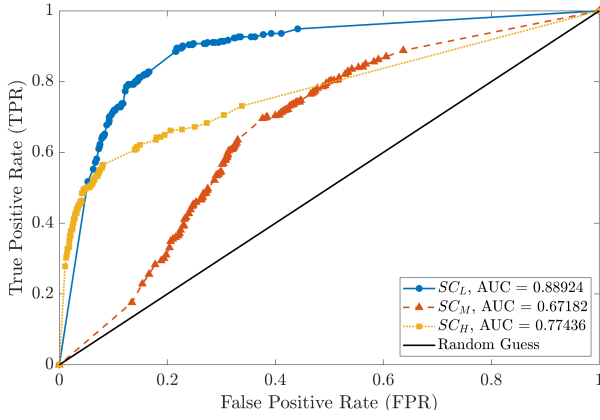
The first reward function is designed to achieve objective 1: maximize performance. Specifically, the objective is to maximize the user’s performance when manually controlling the quadrotor. When the assistance is *off*, a low, medium, or high reward is assigned when the human’s performance is low, medium, or high, respectively. In contrast, the agent is rewarded most for providing automation assistance when the user’s performance is low, and vice versa. To avoid having too many equal reward values, the rewards for medium performance when $a_\theta = \theta_{off}$ are slightly higher than the values when $a_\theta = \theta_{on}$. The resulting optimal policy is also shown in Table III.

TABLE III: Rewards function and action-value functions for achieving objective 1. The highlighted cells represent optimal policy actions.

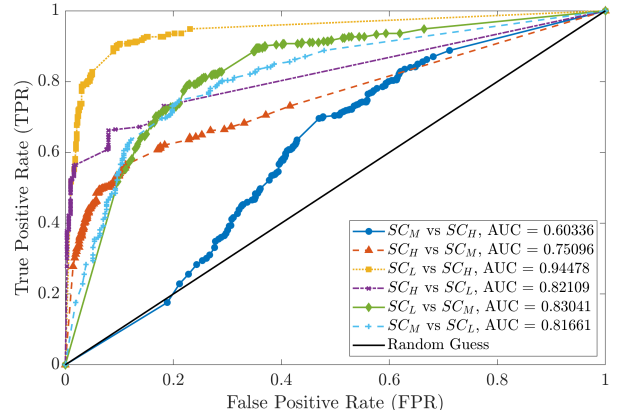
P	SC	$R_1(s, a)$		$Q_1(s, a)$	
		θ_{off}	θ_{on}	θ_{off}	θ_{on}
P_L	SC_L	0.100	0.900	6.29	7.09
	SC_M	0.100	0.900	6.29	7.09
	SC_H	0.100	0.900	6.29	7.09
P_M	SC_L	0.550	0.450	6.74	6.64
	SC_M	0.550	0.450	6.74	6.64
	SC_H	0.550	0.450	6.74	6.64
P_H	SC_L	0.900	0.100	7.09	6.29
	SC_M	0.900	0.100	7.09	6.29
	SC_H	0.900	0.100	7.09	6.29

B. Objective 2: Maximize Self-Confidence

The second reward function is designed to maximize the user’s self-confidence to land the quadrotor manually. This



(a) One-vs-rest ROC curves for self-confidence



(b) One-vs-one ROC curves for self-confidence

Fig. 2: Generated ROC Curves using 80 sets of participant test data.

objective is motivated by existing literature that suggests that increasing self-confidence plays a role in maintaining personal motivation [34], [35], [36], [10] which, in turn, is a part of the humans' *self-efficacy*—the set of beliefs humans have in their own capability to formulate a plan and complete tasks. Therefore, when $a_\theta = \theta_{off}$, the highest reward is assigned when the user's self-confidence state is high, and decreasing awards are assigned to the medium and low self-confidence state. In contrast, the agent is rewarded most for providing automation assistance to the user when their self-confidence is low, and so on. The resulting reward function and optimal policy are shown in Table IV.

TABLE IV: Rewards function and action-value functions to achieve objective 2. The highlighted cells represent optimal policy actions.

P	SC	$R_2(s, a)$		$Q_2(s, a)$	
		θ_{off}	θ_{on}	θ_{off}	θ_{on}
P_L	SC_L	0.100	0.900	5.88	6.67
	SC_M	0.500	0.500	6.12*	6.12
	SC_H	0.900	0.100	6.72	5.92
P_M	SC_L	0.100	1.00	5.88	6.67
	SC_M	0.500	0.500	6.12*	6.12
	SC_H	1.00	0.100	6.72	5.92
P_H	SC_L	0.100	0.900	5.88	6.67
	SC_M	0.500	0.500	6.12*	6.12
	SC_H	0.900	0.100	6.72	5.92

*This value's 4th significant figure is greater.

C. Objective 3: Calibrate Self-Confidence to Performance

The third reward function is designed to calibrate the user's self-confidence to their performance. This objective is based on self-efficacy literature that suggests that if the human is self-aware and realistic about their achievements, they are more likely to experience future success [10], [37]. In the same vein, prior work by the authors has shown that automation assistance intervention designed to calibrate users' self-confidence to performance results in accelerated learning compared to interventions based on only performance [22]. Table V shows the reward function and resulting optimal policy.

TABLE V: Rewards function and action-value functions for achieving objective 3. The highlighted cells represent optimal policy actions.

P	SC	$R_3(s, a)$		$Q_3(s, a)$	
		θ_{off}	θ_{on}	θ_{off}	θ_{on}
P_L	SC_L	0.750	1.00	6.67	6.92
	SC_M	0.750	0.500	6.71	6.45
	SC_H	0.500	0.250	6.47	6.22
P_M	SC_L	0.500	0.750	6.42	6.67
	SC_M	0.750	0.750	6.71	6.70
	SC_H	0.750	0.500	6.72	6.47
P_H	SC_L	0.250	0.500	6.17	6.42
	SC_M	0.500	0.750	6.46	6.70
	SC_H	1.00	0.750	6.97	6.72

Calibration occurs when users' self-confidence and performance levels match. When high self-confidence is calibrated, a high reward of 1.0 is assigned to the agent for $a_\theta = \theta_{off}$, and a slightly lower reward of 0.75 is assigned to the agent if $a_\theta = \theta_{on}$. This is so that the agent is rewarded more when the human is successful at safely landing the quadrotor manually. The agent is rewarded in an opposite manner when users' low self-confidence is calibrated. In other words, a high reward of 1.0 is assigned to the agent if $a_\theta = \theta_{on}$, and a slightly lower reward of 0.75 is assigned to the agent if $a_\theta = \theta_{off}$. This is because when the human's performance is low, they require automation assistance. When medium self-confidence is calibrated, the agent is rewarded a value of 0.75 regardless of a_θ because although self-confidence is calibrated to performance, medium performance is neither the worst case nor best case scenario for the user.

Overconfidence occurs when the user's self-confidence level is higher than their performance level. When the user is overconfident, the agent is rewarded more for not providing automation assistance. In this situation, the user will likely not perform well, leading to self-confidence recalibration to lower performance. There are two levels of overconfidence: light ($s_{SC}, a_P \in \{(SC_H, P_M), (SC_M, P_L)\}$) and severe ($s_{SC}, a_P \in \{(SC_H, P_L)\}$). In the case of light overconfidence, rewards of 0.75 and 0.50 are given to the agent

when $a_\theta = \theta_{off}$ and $a_\theta = \theta_{on}$, respectively. In the case of severe overconfidence, rewards of 0.50 and 0.25 are given to the agent when $a_\theta = \theta_{off}$ and $a_\theta = \theta_{on}$, respectively. The agent is rewarded less when the user is demonstrating severe overconfidence to reduce the miscalibration of self-confidence to performance.

Underconfidence occurs when the user's self-confidence level is lower than their performance level. Similar to overconfidence, there are two levels of underconfidence: light ($(s_{SC}, a_P) \in \{(SC_M, P_H), (SC_L, P_M)\}$) and severe ($(s_{SC}, a_P) \in \{(SC_L, P_H)\}$). If the user is underconfident, the agent is rewarded more for providing automation assistance. This is to help the user improve their performance and increase their self-confidence. This reward-shaping choice also serves to maintain the inverted symmetry of reward values that exists in the reward functions for other objectives for a fairer trained optimal policy comparison. In the case of light underconfidence, rewards of 0.50 and 0.75 are given to the agent when $a_\theta = \theta_{off}$ and $a_\theta = \theta_{on}$, respectively. In the case of severe underconfidence, rewards of 0.25 and 0.50 are given to the agent when $a_\theta = \theta_{off}$ and $a_\theta = \theta_{on}$, respectively.

D. Objective 4: Maximize Performance and Self-confidence.

The fourth and final mode selection policy is designed to achieve objective 4: maximize performance and self-confidence. To do this, we equally weight the reward functions designed for objectives 1 and 2: specifically, $\mathcal{R}_4(s, a_p, a_\theta) = 0.5\mathcal{R}_1(s, a_p, a_\theta) + 0.5\mathcal{R}_2(s, a_p, a_\theta)$, as shown in Table VI. The agent is increasingly rewarded for providing automation assistance as the user's performance decreases. Within each discrete performance level, the agent is rewarded most when the user is in a state of high self-confidence and the automation is off; this is because we want the user to have self-confidence in landing the quadrotor manually.

TABLE VI: Rewards function and action-value functions for achieving objective 4. The highlighted cells represent optimal policy actions.

P	SC	$R_4(s, a)$		$Q_4(s, a)$	
		θ_{off}	θ_{on}	θ_{off}	θ_{on}
P_L	SC_L	0.100	0.900	5.36	6.16
	SC_M	0.300	0.700	5.57	6.96
	SC_H	0.500	0.500	5.82	5.81
P_M	SC_L	0.325	0.625	5.58	5.93
	SC_M	0.525	0.475	5.79	5.74
	SC_H	0.725	0.275	6.04	5.59
P_H	SC_L	0.500	0.500	5.76	5.76
	SC_M	0.700	0.300	5.97	5.56
	SC_H	0.900	0.100	6.22	5.41

E. Verification through Steady-State Analysis

After training optimal policies using the reward functions, steady-state analysis is used to check if, independently of the action taken, all self-confidence states still have a probability of being visited [29]. This ensures that the human will not

be at risk of becoming trapped in isolated loops of self-confidence behavior prior to relaunching the experiment with the optimal policies. The SC MDP model can be recast as a Markov chain by using the trained policies. Therefore, for each trained optimal policy, the transition matrices are reduced from an $|\mathcal{S}| \times |\mathcal{S}| \times |\mathcal{A}|$ matrix to a $|\mathcal{S}| \times |\mathcal{S}|$ matrix. For all policies, the resulting steady-state probabilities show that each state of self-confidence has a possibility of being visited by the human as time step heads towards infinity and that the resulting Markov chains are irreducible.

TABLE VII: Steady-state probabilities.

Policy	$P(SC_L)$	$P(SC_M)$	$P(SC_H)$
Objective 1	0.544	0.295	0.161
Objective 2	0.560	0.160	0.280
Objective 3	0.564	0.282	0.154
Objective 4	0.556	0.285	0.159

V. EXPERIMENTAL RESULTS

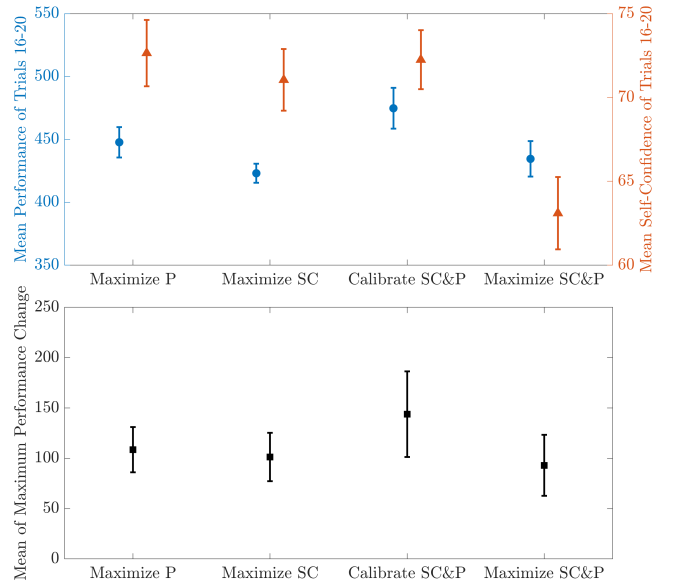


Fig. 3: Top: Mean performance and self-confidence of trials $k = [16, 20]$ for each policy. Bottom: Mean of maximum change in manual performance $\max(\Delta S)$.

The experiment described in Section II was redeployed in closed-loop with each of the four policies designed and synthesized in Section IV. In this between-subjects experiment, we collected 20 sets of participant data for each of the four experimental treatments. Figure 3 depicts the resulting mean performance, mean self-confidence and standard error in the last five trials of all 20 participants from each treatment. Figure 3 also shows the mean of the greatest change in performance of all 20 participants in each policy. For brevity, the performance and self-confidence of the last five trials will be denoted as $S_{k=[16,20]}$ and $SC_{k=[16,20]}$, respectively. The maximum change in performance is calculated as $\max(S_{k=[16,20]}) - S_{k=2}$ and is denoted by $\max(\Delta S)$.

Recall that in each experimental treatment, the first two and last five trials are completed in manual mode such that final performance and change in performance can be quantified. The second trial is chosen as the baseline for comparison since the first trial serves as a practice round for participants. It should be noted that the range of scores in Figure 3 corresponds to unsuccessful and unsafe landings. Although the range does not span the entire score space, performance improvement is still captured.

A one-way ANOVA test is applied to the $S_{k=[16,20]}$ and $SC_{[16,20]}$ data, as well as all participant $\max(\Delta S)$ to determine whether there are statistically significant differences between treatments. The difference in means between all four groups of participants was found to be statistically significant for $S_{k=[16,20]}$ ($F(3, 396) = 2.97, p = 0.0314$) and $SC_{k=[16,20]}$ ($F(3, 396) = 5.37, p = 0.00123$). The difference in means for $\max(\Delta S)$ was not found to be statistically significant ($F(3, 76) = 5.26, p = 0.666$).

VI. DISCUSSION

From the experimental results and statistical analysis presented in Section V, we can draw the following conclusions.

A mode selection policy based on cognitive feedback leads to statistically significant improvements in performance compared to one that only considers performance feedback. The control policy designed to calibrate self-confidence to performance, $Q_3(s, a)$, resulted in the best mean $S_{k=[16,20]}$ ($\mu = 475$), second highest $SC_{k=[16,20]}$ (72 out of 100) and the highest mean $\max(\Delta S)$ among all 80 participants. In comparison, the participants who experienced the treatment that implemented $Q_1(s, a)$ (maximizing performance) achieved a lower mean $S_{k=[16,20]}$ ($\mu = 448$) and lower mean $\max(\Delta S)$. Interestingly, the $Q_1(s, a)$ treatment participants reported a similar high $SC_{k=[16,20]}$ (73 out of 100) to the $Q_3(s, a)$ treatment participants (72 out of 100). However, upon closer examination, this self-confidence rating does not match the level of performance achieved by $Q_1(s, a)$ treatment participants. One explanation for this is that the $Q_1(s, a)$ policy, shown in Table III, is skewed towards not providing the participant with automation assistance. This means that participants are achieving at least medium performance, leading to $a_\theta = \theta_{off}$, and in response, participants' self-confidence is higher.

The control policy designed to maximize self-confidence, $Q_2(s, a)$, resulted in similar high mean $SC_{k=[16,20]}$ (71 out of 100) to $Q_3(s, a)$ and the lowest mean $S_{k=[16,20]}$ ($\mu = 423$). This shows that the policy is successful at maximizing self-confidence, however, the self-confidence rating is not appropriate for the level of performance.

Calibrating self-confidence to performance is more effective in improving learning outcomes compared to maximizing both performance and self-confidence. The $Q_4(s, a)$ treatment participants (maximize both performance and self-confidence) achieved the second lowest mean $S_{k=[16,20]}$ ($\mu = 435$) and mean $SC_{k=[16,20]}$ (63 out of 100) as well as the lowest mean $\max(\Delta S)$ among all 80 participants. In contrast, when training the more successful optimal control

policy that calibrated self-confidence to performance, a reward was assigned for each case of performance action, self-confidence state, and automation assistance action. In other words, the specific interaction between performance and self-confidence should be considered when incorporating both cognitive and performance feedback in a cognitive control policy.

VII. CONCLUSION

We designed and validated multiple model-based cognitive control policies, motivated by the important role of learners' self-confidence while learning, to improve learning outcomes in a psychomotor task. In particular, we applied reward shaping to design and then compare four different policies that aimed to achieve well-established learning objectives related to learners' performance and self-confidence during the task. Users who experienced a policy designed to calibrate self-confidence to performance achieved the highest mean score across the last five trials among the four policies considered. Notably, the results suggest that the best cognitive control objective is one that aims to *calibrate* learners' self-confidence to their performance, as compared to a more naive approach of only maximizing performance, or maximizing self-confidence. Future work will consider different metrics to quantify performance or learning, improving the automation assistance by having different extents to which it helps the user or adapting the control policy to the learning stage the learner is in. Additional data collection would also enable increased resolution of the model's discretized states and actions.

ACKNOWLEDGMENT

We thank Sooyung Byeon (Purdue University) for the initial development of the game platform.

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